Getting Started

The steps below will install and configure ELMFIRE. This has been tested on a clean install of Ubuntu Server 24.04.1 LTS. Modifications to run on other Linux distributions may be required.

Install prerequisites

The following commands will install packages needed to build and run ELMFIRE:

```
sudo apt-get update && sudo apt-get upgrade -y
sudo apt-get install -y bc csvkit gdal-bin gfortran git jq libopenmpi-dev \
                        openmpi-bin pigz python3 python3-pip unzip wget zip
```

Several Python libraries / packages, including three related to Google Remote Proecedure Call (gRPC), are needed to run the CloudFire microservices that provide ELMFIRE with fuel, weather, and ignition geospatial data. These can be installed system-wide as follows:

```
sudo pip3 install google-api-python-client grpcio grpcio-tools python-dateutil \
                  --break-system-packages
```

Alternatively, *virtualenv* can be used to prevent forcing a system-wide install as follows:

```
sudo apt-get install python3-virtualenv
virtualenv $HOME/virtualenv/elmfire
source $HOME/virtualenv/elmfire/bin/activate
/$HOME/virtualenv/elmfire/bin/pip3 install google-api-python-client grpcio grpcio-
tools \
                                           python-dateutil
```

Clone ELMFIRE Github repository

The current ELMFIRE repository can be cloned as follows:

```
git clone https://github.com/lautenberger/elmfire.git
```

Since this will clone the current repository, including all recent commits, for use in production environments a user may want to clone the latest stable release / branch instead, i.e.:

```
git clone --branch 2025.0212 --single-branch
https://github.com/lautenberger/elmfire.git
```

Set environment variables

ELMFIRE uses four environment variables:

1. `ELMFIRE_SCRATCH_BASE`: Full path to a scratch directory where the user has read/write access.
2. `ELMFIRE_BASE_DIR`: Full path to the directory to which the ELMFIRE Git repository was cloned (i.e., the directory containing `README.md`).
3. `ELMFIRE_INSTALL_DIR`: Full path to the directory where the ELMFIRE executable files will be installed (default: `$ELMFIRE_BASE_DIR/build/linux/bin`).
4. `CLOUDFIRE_SERVER`: IP address of Cloudfire microservices server (this should be set to `worldgen.cloudfire.io`, more on this later).

The easiest way to specify these variables is by exporting them from your `~/.bashrc` file. For example, one could `pico ~/.bashrc` and then add the following lines (replacing `/scratch/clauten`, `/home/clauten/elmfire`, etc.):

```
export ELMFIRE_SCRATCH_BASE=/scratch/clauten
export ELMFIRE_BASE_DIR=/home/clauten/elmfire
export ELMFIRE_INSTALL_DIR=$ELMFIRE_BASE_DIR/build/linux/bin
export CLOUDFIRE_SERVER=worldgen.cloudfire.io
export PATH=$PATH:$ELMFIRE_INSTALL_DIR:$ELMFIRE_BASE_DIR/cloudfire
```

These environment variables will be set on the next login or after sourcing the newly-edited file (`. ~/.bashrc`).

Build ELMFIRE executables

ELMFIRE and its postprocessing tool can be built as follows:

```
cd $ELMFIRE_BASE_DIR/build/linux
./make_gnu.sh
```

Unless an error occurs, this will build the executables `elmfire_VERSION` and `elmfire_post_VERSION` (where version is, for example, 2025.0212) and copy them to `$ELMFIRE_INSTALL_DIR`. If this directory is not in the user's `$PATH` it should be added at this time. Note that two debug executables are also built.